# Sandbags for Home Runoff

Fill sandbags efficiently using a multi-bag filling fixture.

## BEFORE YOU BEGIN

Sandbags can redirect shallow stormwater runoff from vulnerable areas of your property. They work best when placed before heavy rain arrives.

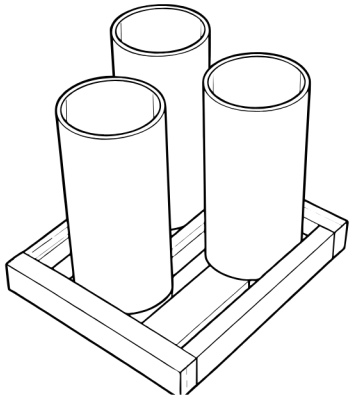
## WHAT YOU'LL NEED

- Empty woven sandbags
- Clean sand or suitable granular fill
- Multi-bag filling fixture
- Shovel
- Gloves and sturdy footwear
- Eye protection when sand is dusty
- A clear, level work area
- A partner for lifting

> **△ CAUTION — PREVENT HARM**
>
> A filled bag may weigh 35–40 lb. Use a partner or lighter bags if needed. Work on level, dry ground; keep hands and feet clear when lifting the fixture. Leave bags one-half to two-thirds full, with fabric to fold underneath; remove excess sand.

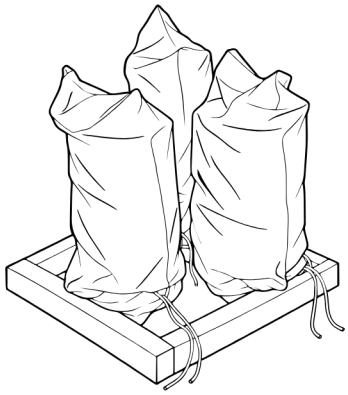
## FILL SANDBAGS WITH THE FIXTURE

**1** **Invert the fixture.**



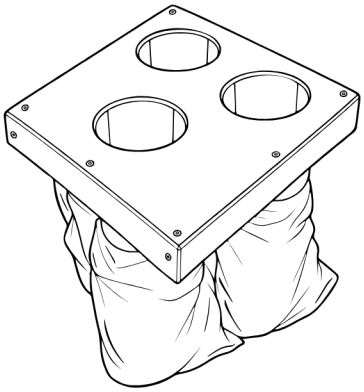
1. Invert the fixture.

**2** **Put a sandbag on each pipe.**



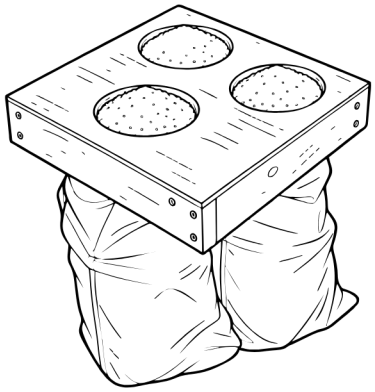
2. Put a sandbag on each pipe.

**3** **Turn the fixture right side up.**



3. Turn the fixture right side up.

**4** **Fill the fixture openings with sand.**



4. Fill to the top with sand.

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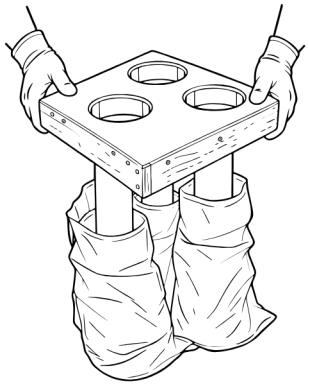
# Sandbags for Home Runoff — continued

Remove filled bags, shape them for placement, and store them safely.



## COMPLETE THE FILLING CYCLE

**5** **Lift the fixture and leave the sand in the bags.** Use a partner and lift together.



5. Lift straight up. Let the filled bags remain on the ground.

**6** **Set the fixture upside down.**



6. Reset the fixture for more sandbags.

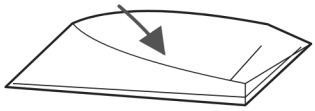
**7** **Move the sandbags away.**



Four sandbags lying in a row.

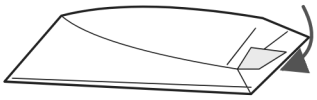
## SHAPE AND FOLD THE BAGS

**8** **Flatten sand toward the closed end.**



8. Flatten the sand toward the closed end.

**9** **Fold the empty end underneath.** Do not tie barrier bags.



9. Fold the empty end underneath.

## STORE FILLED SANDBAGS

- Stack low on a pallet or stable support.
- Protect from sun and weather.
- Keep driveways and exits clear.
- Inspect for damage before use.



Sandbags should rotate 90-degrees at each layer to interleave like bricks with the bags below

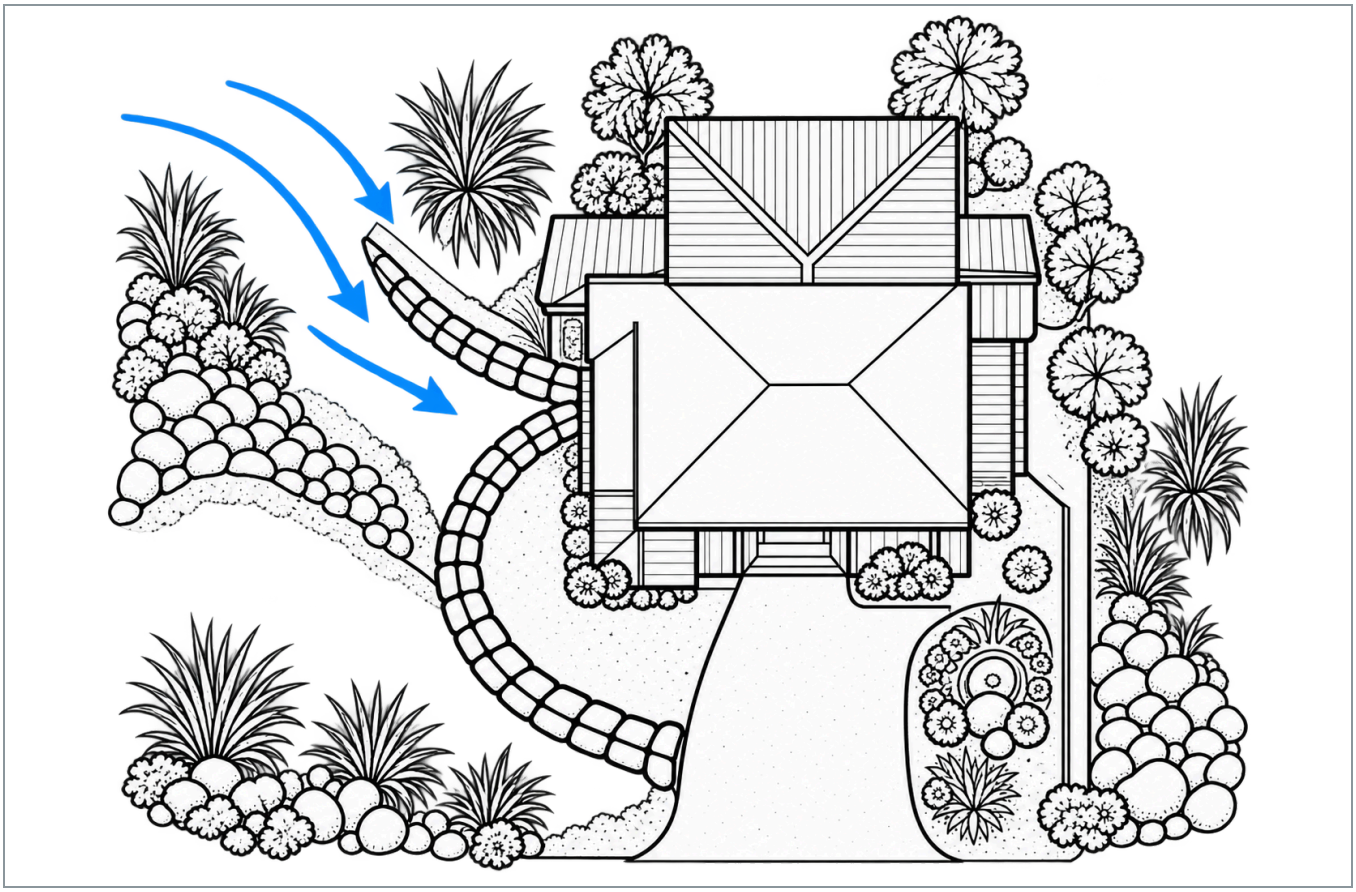
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# Planning a Sandbag Barrier

Understand where water comes from and where it can safely go.

## FIRST, OBSERVE THE WATER'S PATH

1. **Where is the water coming from?** Look for runoff from hillsides, driveways, roofs, and low areas.
2. **What needs protection?** Identify vulnerable doors, garages, low windows, and openings.
3. **Where can the water safely go?** Find an existing drainage path on your property.



Use sandbags to redirect water to a safe outlet. Do not dump your water into your neighbors lawn.

## SELECT THE BARRIER LOCATION

- Place bags on firm, level ground.
- Intercept runoff before it reaches the opening.
- Use a short line that redirects water.
- Keep drains, exits, and access routes open.
- Do not redirect runoff toward a neighbor, road, or another doorway.

**× DANGER — STOP NOW**

Leave if water is moving fast, rising around you, or cutting off your exit. Do not enter flowing water or build beside an unstable slope or wall.

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# Building a Sandbag Barrier

Overlap bags, stagger joints, and maintain a stable base.

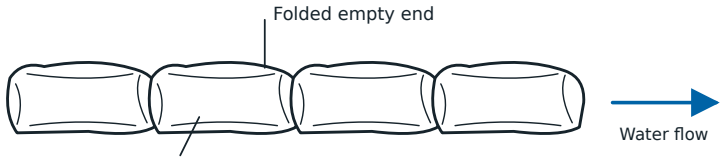
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## PREPARE THE GROUND

1. **Clear the area.** Remove loose debris and fill low spots.
2. **Establish the line.** Identify the water-facing side and intended direction of flow.

## PLACE THE FIRST COURSE

3. **Lay bags flat.** Fold the empty end underneath and toward incoming water.
4. **Overlap adjoining bags.** Lay the filled part over the previous bag's empty end.
5. **Seat the bags.** Press or tamp firmly to reduce gaps.



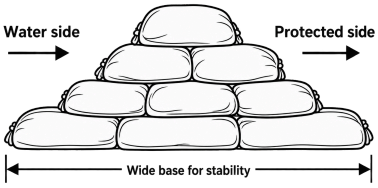

Top view — overlap adjoining bags.

## ADD ADDITIONAL COURSES

6. **Stagger every course.** Offset joints by about half a bag.
7. **Maintain a wide base.** Keep the barrier low with a stable, stepped profile.
8. **Check the ends.** Connect to suitable higher ground or provide a safe outlet.



Sandbags staggered around a basement window.




As you increase the height of a dyke, increase the width of the dyke as well.

## INSPECT THE FINISHED BARRIER

- Check for gaps, uncovered joints, and displaced bags.
- Look for water flowing around the ends or pooling behind the line.
- Sandbags are temporary and do not form a watertight seal.

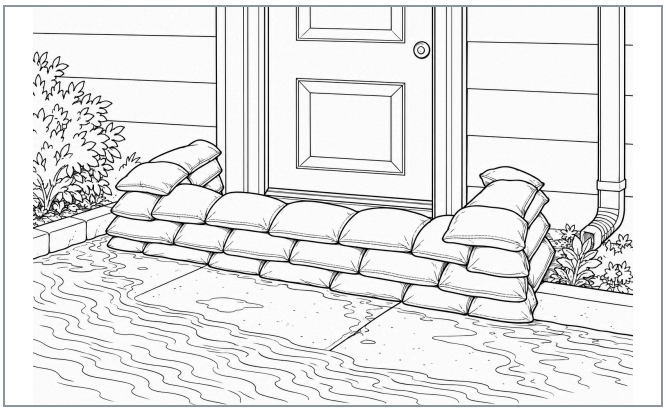
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# Protecting Vulnerable Openings

Choose the barrier arrangement that fits the runoff problem.

## A. EXTERIOR DOORWAYS

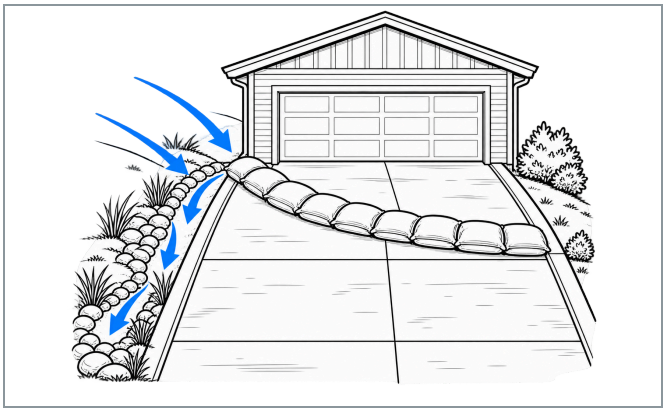
- Place a short, low barrier before water reaches the threshold.
- Extend it beyond both sides of the opening.
- Avoid blocking the only safe exit.
- Check for water entering around the ends or through other openings.
- Do not rely on sandbags to make a doorway watertight.



Doorway protection. Keep another safe exit available.

## B. GARAGE DOORS AND SLOPING DRIVEWAYS

- Intercept runoff uphill of the garage door.
- Use an angled or curved line to redirect water.
- Lead water to an existing suitable drainage path.
- Avoid directing water toward a neighbor, road, or other doorway.
- Keep the evacuation route open.

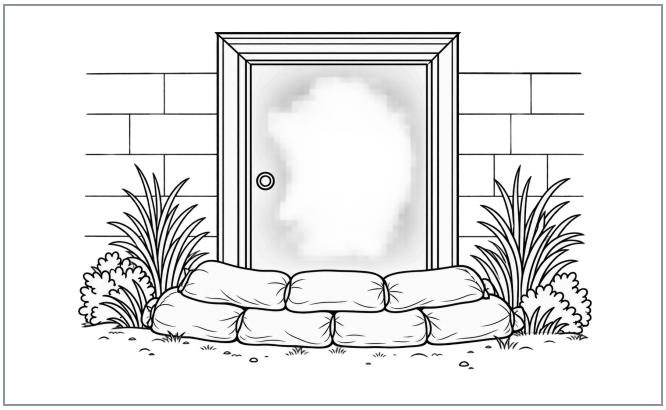


Use sandbags to divert water flow along safe paths. If it overtops the sandbags, give it a safe location for the water to flow.

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## C. LOW WINDOWS AND UTILITY OPENINGS

- Identify low windows, window wells, and utility penetrations.
- Use a low diversion barrier to reduce surface runoff.
- Keep required ventilation, window escape routes, and utility access clear.



If you place sandbags around a door, give yourself enough room to enter and exit.

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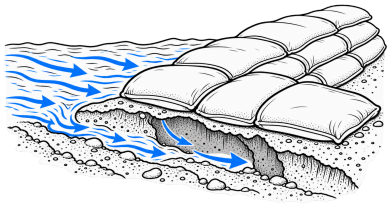
# Understanding Water Movement

Recognize bypass, seepage, and unintended pooling.



## 1. WATER FLOWING AROUND THE ENDS

A line that stops below higher ground may send water around its end.



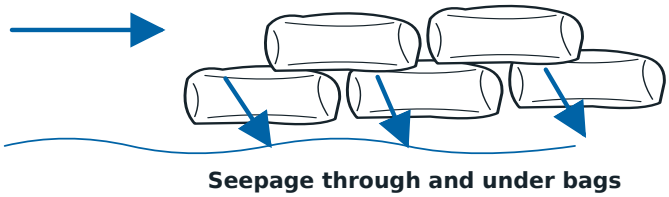
Fast moving water is undermining a sandbag installation

### What to do

- Extend or adjust the line only from safe, dry ground.
- Use suitable higher ground or an existing safe outlet.

## 2. WATER SEEPING THROUGH THE BARRIER

Water can pass between bags, through fabric, and underneath.



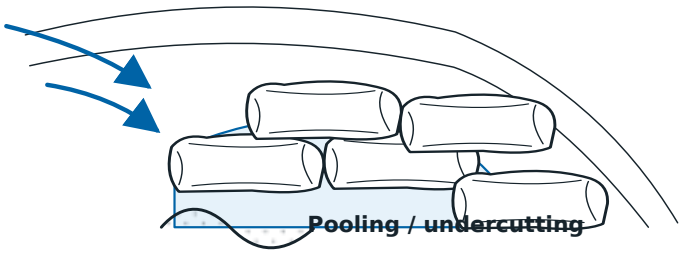

Water seeping through the barrier.

### What to do

- Seat bags firmly and cover joints.
- Follow local guidance for water-side plastic sheeting.
- Inspect for leakage from safe ground.

## 3. WATER POOLING BEHIND THE BARRIER

A poorly placed barrier can trap water on the protected side.




Water pooling behind the barrier.

### What to do

- Do not work in flowing water or on unstable ground.
- Move to safety if erosion, displacement, or a rapid rise occurs.

**× DANGER — STOP NOW**

Leave if conditions become unsafe. Undercutting, displaced bags, or rising water can cause the barrier to fail.

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# Sandbag Mistakes to Avoid

Six common errors that reduce protection.

## 1. Aligned vertical joints

× WRONG



Aligned joints

✓ RIGHT

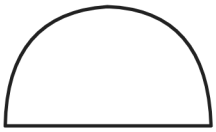


Offset joints

bag joints line up vertically. the upper bag bridges the joint below.

## 2. Overfilled bags

× WRONG



Overfilled and rounded

✓ RIGHT



Partly filled and flexible

an overfilled rounded bag cannot settle tightly. a partly filled bag lies flat and remains flexible.

## 3. Tall, narrow stacks

× WRONG



Tall, narrow base

✓ RIGHT

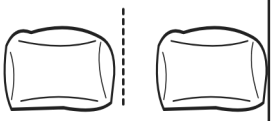


Wide, stepped base

four bags form a narrow column. a low stepped stack has a broad stable base.

## 4. Open gaps and loose ends

× WRONG



Gaps and loose ends

✓ RIGHT

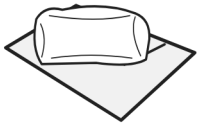


Overlap and fold tails

separated bags leave a gap. flattened bags overlap with empty ends folded underneath.

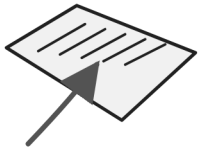
## 5. Blocking drainage

× WRONG



Drain blocked

✓ RIGHT

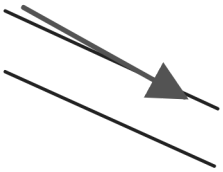


Keep drains open

a sandbag blocks a drain grate. the drain is clear, with a labeled runoff arrow entering it.

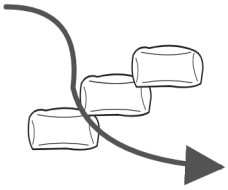
## 6. Unsafe diversion

× WRONG



Toward neighbor / road

✓ RIGHT



To a safe outlet

runoff arrows point toward a neighbor or road. a low line steers water to an existing safe outlet on the property.

**× DANGER — STOP NOW**

Keep exits open. Do not work in moving water or build beside an unstable wall or slope. Leave if conditions become unsafe.

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# Sandbag Field Checklist

Prepare early, inspect safely, and remove temporary barriers when conditions permit.

## BEFORE THE STORM

- ☐ Identify vulnerable openings and runoff paths.
- ☐ Locate a safe, functioning drainage route.
- ☐ Clear accessible drains and drainage channels.
- ☐ Obtain sandbags, sand, and necessary tools.
- ☐ Fill and stage bags on stable ground.
- ☐ Build the diversion before water arrives.
- ☐ Check that exits remain accessible.

## DURING THE STORM

- ☐ Observe runoff from a safe location.
- ☐ Watch for water bypassing the barrier.
- ☐ Check for unexpected pooling or seepage.
- ☐ Stay away from moving water and unstable slopes.
- ☐ Stop work and leave if conditions become dangerous.



Inspect from safe ground; keep exits open.

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## AFTER THE STORM

- ☐ Wait until conditions are safe before approaching.
- ☐ Inspect for erosion, unstable ground, and displaced bags.
- ☐ Remove temporary barriers when no longer needed.
- ☐ Do not dump used sand into drains or natural channels.
- ☐ Follow local disposal guidance for wet or contaminated bags.
- ☐ Correct drainage problems before the next storm.

**△ CAUTION — PREVENT HARM**

Sandbags redirect and slow water; they do not make a home floodproof. Do not enter floodwater, approach unstable slopes, or stay in a threatened structure to maintain a barrier.

**SOURCES**

**U.S. Army Corps of Engineers, St. Paul District.** *Flood Fight Handbook* (2022), §§2.1–2.3. Flood Fight Handbook PDF.
**RiverOakRanch.** “Filling Sandbags the Easy Way, With Sandbag Tubes” (2010), Instructables. Fixture instructions. Both sources accessed 2026-09-24.